

Mohamed Islam Elshourbagy

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PROFESSIONAL SUMMARY

Computer Science & AI student at Alamein International University with hands-on experience building intelligent systems, NLP pipelines, and full-stack AI applications. Proficient in Python, machine learning, deep learning, and REST API development, with practical exposure to containerized deployment, graph algorithms, and real-time systems. Demonstrated ability to deliver complete projects end-to-end from data preprocessing and model training to frontend integration and deployment. Seeking ML engineering roles to apply AI expertise in production environments.

EDUCATION

BSc Computer Science — Artificial Intelligence

Oct 2022 – Present

AI Alamein International University (AIU), Egypt

- Specialization in Artificial Intelligence covering machine learning, deep learning, NLP, algorithms, and software engineering.

EXPERIENCE

Agentic AI & Generative AI System Developer

Jul 2026 – Present

Digital Egypt Pioneers Initiative (DEPI), Egypt — Hybrid

- Selected for DEPI, a government-backed national technical training initiative focused on building advanced digital skills for the Egyptian tech workforce.
- Currently developing hands-on expertise in agentic AI system architecture, LLM-based application design, and generative AI development workflows.
- Engaging with practical training covering AI agent orchestration, prompt engineering, and applied machine learning concepts as part of an intensive, project-based curriculum.
- Building foundational proficiency in tools and frameworks relevant to generative AI development (e.g., LangChain, RAG pipelines, LLMS).

Game Development & Animation Intern

Jul 2025 – Sep 2025

City of Scientific Research and Technological Applications (SRTA-City), Egypt — Hybrid

- Developed 3D character models with complete skeletal rigging in Blender, producing fluid animations and realistic movement for in-game characters.
- Implemented AI agent behaviors using Unity's NavMesh system for intelligent pathfinding and navigation, creating NPCs that dynamically respond to environmental obstacles and player actions.
- Applied character rigging, AI agent behavior programming, and animation pipeline workflows for interactive game development in Unity.
- Built and textured 3D assets using advanced modeling techniques, ensuring performance-optimized assets suitable for real-time rendering.

Software Development Trainee

Jul 2024 – Aug 2024

Information Technology Institute (ITI), Egypt — Hybrid

- Developed a full-featured Chat Room Application in Java (JavaFX, FXML, Swing) with user authentication, real-time messaging, and active status indicators.
- Implemented multi-threaded server architecture using Java socket programming to handle simultaneous user connections, improving system reliability and performance.
- Designed responsive and intuitive user interfaces with JavaFX and FXML, applying UI/UX best practices to enhance the user experience.

- Applied concurrent programming and multi-threading concepts to support real-time data processing across connected clients.

PROJECTS

Efficient Sentiment Classification using GPT-2, LoRA & Knowledge Distillation | *Python, PyTorch, HuggingFace Transformers, LoRA (PEFT)*

- Designed a parameter-efficient NLP pipeline combining Low-Rank Adaptation (LoRA) with Teacher-Student Knowledge Distillation for sentiment classification, training only **2.15% of total parameters** while retaining competitive performance.
- Fine-tuned a full GPT-2 model as a teacher and distilled its knowledge into a LoRA-adapted DistilGPT-2 student using KL-divergence loss with label smoothing and Focal Loss for class imbalance.
- Engineered a hybrid pooling strategy (80% attention pooling, 20% mean pooling) replacing standard finaltoken extraction, producing richer sentence-level embeddings for the classification head.
- Achieved **0.8962 Macro F1** on the IMDB dataset (vs. teacher's 0.9245) and evaluated on SST-5 fine-grained 5-class sentiment, demonstrating strong generalization under resource constraints.
- Applied Mixed Precision Training, Gradient Accumulation, Gradient Checkpointing, and Cosine LR Scheduling for training stability and efficiency.

Intelligent Cairo Transportation System | *Python, Scikit-Learn, Flask, Docker, Leaflet.js*

- Developed a containerized full-stack web application to visualize and compare intelligent transport planning and routing algorithms across Greater Cairo.
- Engineered a machine learning-assisted traffic prediction module using Scikit-Learn and NumPy to power time-dependent, adaptive pathfinding.
- Implemented advanced graph routing (A*, Dijkstra) and dynamic programming algorithms to optimize emergency vehicle routing, bus allocation, and maintenance budgets.
- Designed and exposed RESTful API endpoints using Flask to seamlessly integrate backend optimization logic with an interactive Leaflet map UI.

COURSES & CERTIFICATIONS

Neural Networks and Deep Learning — DeepLearning.AI | Coursera coursera.org/verify/H10PV0ENSIYL

NLP with Classification and Vector Spaces — DeepLearning.AI | Coursera coursera.org/verify/27WE0B5L89X0

NLP in TensorFlow — DeepLearning.AI | Coursera coursera.org/verify/SM9I2M78KLMY

Building Multi-Agent Systems — Microsoft | Coursera coursera.org/verify/1L0A6BPBW69S

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, SQL, HTML, CSS, R, C#

Frameworks & Libraries: Scikit-Learn, PyTorch, TensorFlow/Keras, HuggingFace Transformers, NumPy, Pandas, Matplotlib, NLTK, spaCy, Flask, Streamlit, SymPy, JavaFX, Unity (NavMesh)

Tools & Platforms: Docker, Git/GitHub, VS Code, PyCharm, IntelliJ IDEA, ModelSim, Blender, Weka, RStudio

Concepts & Domains: Machine Learning, Deep Learning, NLP, Large Language Models (LLMs), ParameterEfficient Fine-Tuning (LoRA), Knowledge Distillation, LSTM Networks, Graph Algorithms (A*, Dijkstra), REST APIs, AI Agent Programming, Multi-threading, Socket Programming, OOP

Operating Systems: Windows, Linux

Soft Skills: Leadership, Analytical Thinking, Problem-Solving, Client Communication, Time Management

LANGUAGES

Arabic: Native | **English:** Proficient